Title: EXAM READINESS AND PERFORMANCE FORECASTER

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Abstract

- EduPulse is a pure frontend web app that forecasts student exam readiness using real academic data
- Built entirely with HTML, CSS, and JavaScript — no backend, no frameworks, no external libraries
- Students input attendance %, topic-level syllabus completion, and exam date across multiple subjects
- Readiness score computed using weighted formula: Attendance (35%) + Syllabus (45%) + Time Factor (20%)
- Topic-level sliders allow granular completion tracking per subject
- Provides live exam countdown (setInterval), subject-wise scores, and readiness level: High / Medium / Low
- Personalized study recommendations generated per subject based on calculated score
- All data persists across sessions using localStorage — no server or database required

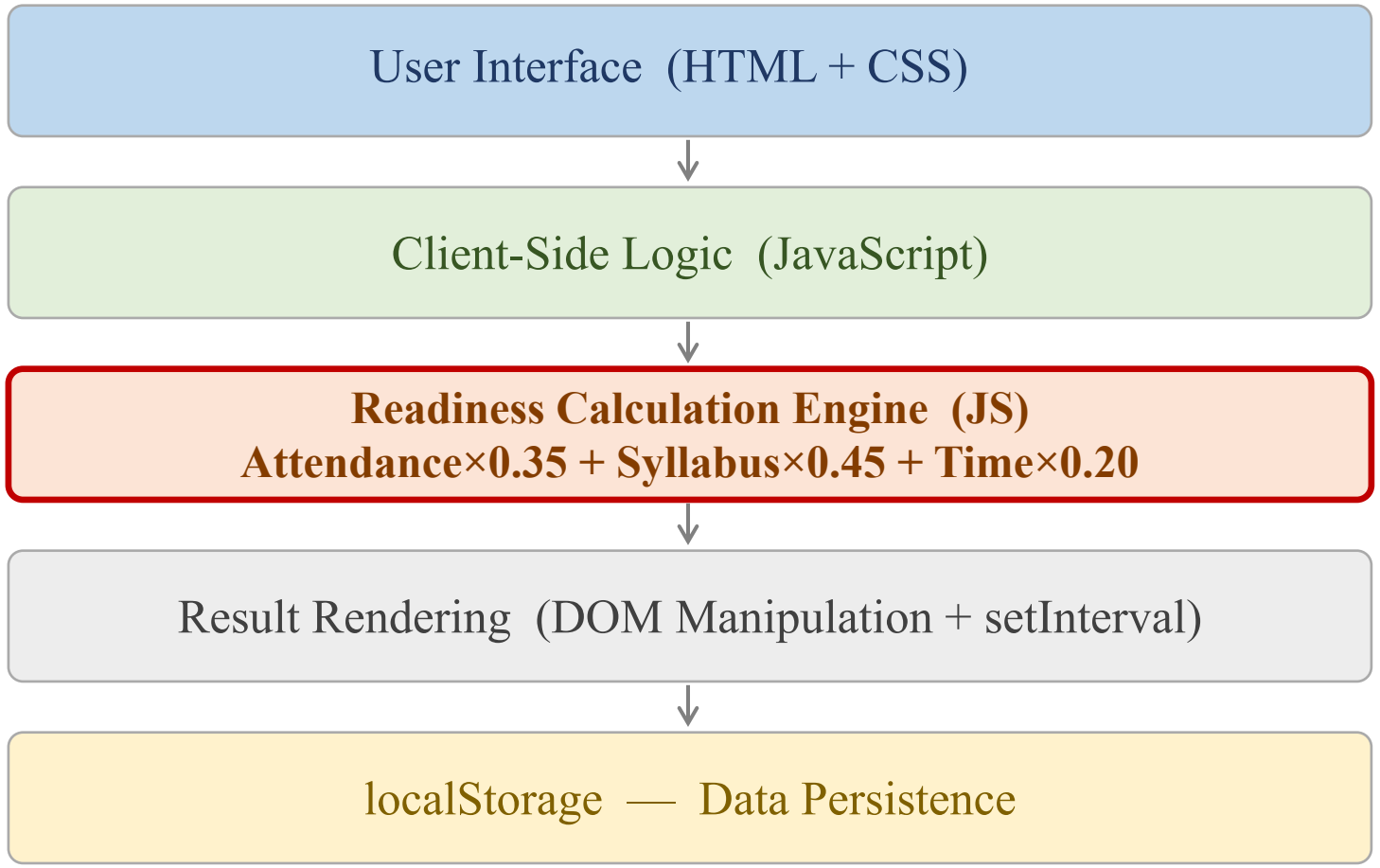
Existing System

- Students manually estimate exam preparation with no structured tracking method
- Attendance and syllabus progress tracked separately using spreadsheets or paper
- No unified tool to combine multiple parameters into a single readiness score
- Study planning based on intuition rather than data-driven insights
- No real-time awareness of time remaining relative to preparation level
- Generic advice from peers or teachers — not personalized to individual progress

Proposed System

- EduPulse consolidates attendance, syllabus completion, and time factor into one browser-based platform
- Weighted readiness formula produces an objective, personalized score per subject
- Topic-level sliders capture granular syllabus progress beyond simple percentage inputs
- Live exam countdown (setInterval) creates urgency awareness tied directly to readiness level
- Automated subject-wise recommendations guide focused, priority-based study sessions
- Client-side JavaScript applies all logic — works offline, no installation, no server required
- Fully browser-based with localStorage persistence — zero setup or external dependency

System Architecture



Project Modules

Authentication Module
Login/signup with validation, localStorage user storage, tab switching

Exam Setup Module
Exam date picker, subject count input, date validation

Subject Input Module
Dynamic subject name cards, topic entry via textarea

Data Entry Module
Attendance input, per-topic range sliders.

Readiness Engine
Weighted formula: Attendance×0.35 + Syllabus×0.45 + Time×0.20

Result Display Module
Overall score card, per-subject cards, readiness level badges

Countdown Module
setInterval-based live countdown to exam day in days and hours

Persistence Module
localStorage save/restore for progress across sessions

Objectives

- To build a fully browser-based student readiness tool with zero external dependencies
- To apply a weighted multi-parameter formula to objectively forecast exam preparedness
- Enabling topic-level syllabus tracking through interactive sliders per subject
- Implementing client-side user authentication using localStorage
- To provide automated, score-based study recommendations per subject
- To demonstrate JavaScript DOM manipulation, event handling, and setInterval
- Using semantic HTML5 and responsive CSS (Flexbox + Grid) for accessible UI design
- Persist student progress across browser sessions without any backend or database

Co’s Attainments

CO	Topic Area	Implementation in Exam Readiness And Performance Forecaster
CO1	HTML & Introductory CSS	Semantic tags (header, nav, main, section, article, aside, footer, figure, fieldset, legend, hgroup), meta tags, external stylesheet, Google Fonts, CSS custom properties
CO2	Forms, Layout & Responsiveness	fieldset + legend, CSS Grid (auth-wrapper, metrics-grid), Flexbox (all panels), transitions, animations (@keyframes), pseudo-classes (:hover, :focus, :nth-child), pseudo-elements (::before, ::after, ::placeholder), attribute selectors, media queries (768px, 480px)
CO3	JavaScript Essentials	Functions, objects (app state), for loops (no array methods used), arrays, if/else, push, break — all logic in auth.js and dashboard.js
CO4	DOM & Events	Dynamic DOM building (buildSubjectNameInputs, buildSubjectDataInputs, displayResults), onclick/onsubmit handlers, setInterval countdown, localStorage CRUD, step navigation
CO5	SEO & Validation	Meta description, keywords, author tags; input types (email, password, date, number, range, text); required, minlength attributes; ARIA roles and labels for accessibility

Co's Attainments | CO1

CO1: Internet Fundamentals, HTML & Introductory CSS

HTML Document structure: <!DOCTYPE>, <html lang>, <head>, <body>

Meta tags: charset, viewport, description, keywords, author — SEO implementation

Heading hierarchy: h1 (brand name), h2 (slide titles), h3 (card headers)

Semantic tags: <header>, <nav>, <main>, <section>, <article>, <aside>, <footer>, <figure>

Form elements: <form>, <fieldset>, <legend>, <label>, <input>, <textarea>, <button>

Links: <a href> for tab switching and page redirect on logout

CSS basics: selectors, properties, values, external stylesheet linked via <link rel="stylesheet">

CSS custom properties (variables) in :root for colors, spacing, radius, shadow

Typography: Google Fonts imported (Cormorant Garamond + Plus Jakarta Sans), font-size, font-weight, line-height

Co's Attainments| CO2

CO2: HTML Forms, CSS Layout & Responsiveness

Forms: input types — email, password, text, number, date, range; attributes — required, minlength, placeholder, autocomplete

fieldset + legend groups related inputs; hgroup pairs heading with subtitle

CSS Grid: auth-wrapper (2-column login layout), metrics-grid (3-column results)

Flexbox: auth-left, auth-right, tab-bar, header-nav, steps-bar, btn-row, feature-list

Transitions: border-color, box-shadow on focus; transform on hover; background on tab switch

Animations: @keyframes fadeUp on .dash-card — slides up on step change

Pseudo-classes: :hover (links, buttons, list items), :focus (inputs), :nth-child (result cards), :disabled, :not()

Pseudo-elements: ::placeholder (italic hint text), ::before (gold dot bullets), ::after (brand-name underline)

Attribute selectors: input[type="email"], input[type="range"], input[type="date"] etc.

Media queries: 768px (single column auth), 480px (stacked buttons, hidden step labels)

Co's Attainments | CO3 & CO4

CO3: JavaScript Programming Essentials

- Functions: showTab(), handleLogin(), handleSignup(), analyzeReadiness(), getTimeScore(), getLevel(), getRecommendation()
- Objects: app{} stores all dashboard state — currentStep, examDate, subjects, subjectData, countdownTimer
- For loops: all iteration uses for loops — no .map(), .filter(), .reduce(), .find() used anywhere
- Arrays: push() to add subjects/topics; manual index-based access throughout
- If/else: form validation, score classification (High/Medium/Low), time factor assignment

CO4: JavaScript Interactivity, DOM & Events

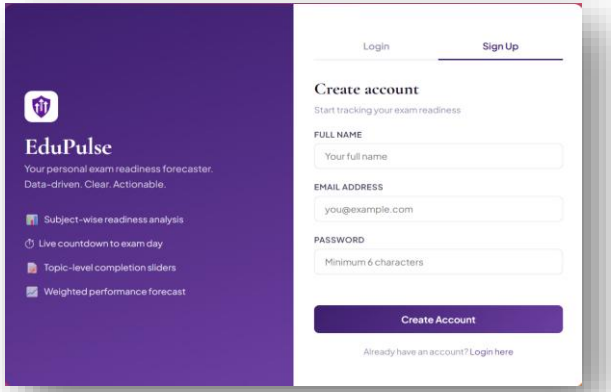
- DOM manipulation: createElement, appendChild, innerHTML, getElementById — buildSubjectNameInputs(), buildSubjectDataInputs(), displayResults()
- Event handling: onsubmit (forms), onclick (buttons, tabs, sliders), oninput (range sliders)
- setInterval: updateCountdown() runs every second, updates days/hours DOM in real time
- localStorage: get/set/remove for edupulse_users, edupulse_loggedin, edupulse_progress

Co's Attainments | CO5

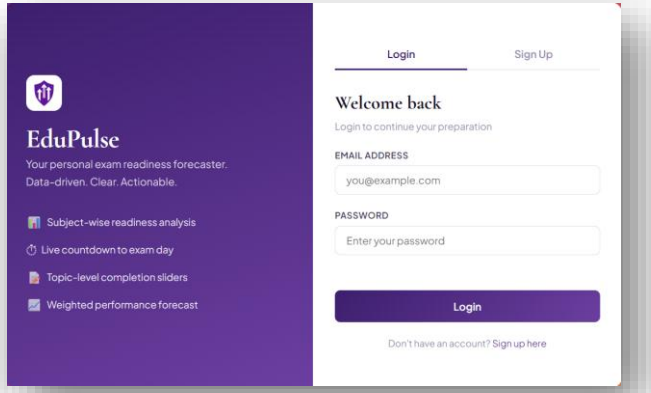
CO5: SEO & Advanced Form Handling

- Meta tags: description, keywords, author for search engine optimization
- ARIA attributes: role="tablist", aria-selected, aria-controls, aria-label, aria-live="polite" for accessibility
- Input validation attributes: required, minlength, type="email/date/number" enforce data correctness
- Form error feedback: <p role="alert" aria-live="polite"> dynamically updated by JavaScript

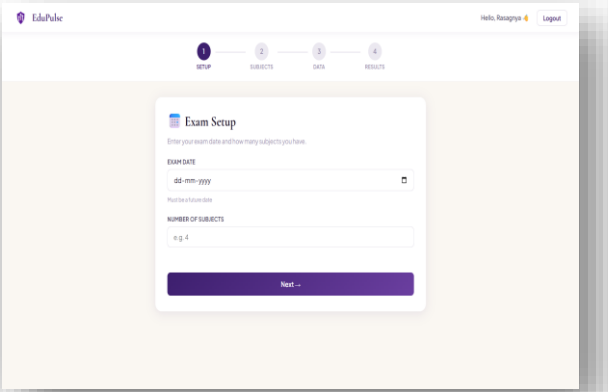
Screenshots



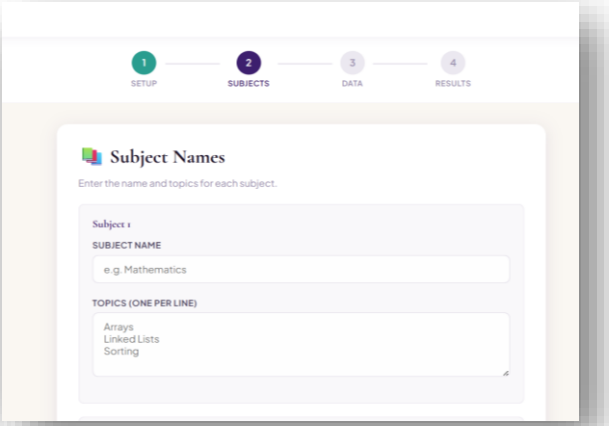
SIGN UP Page



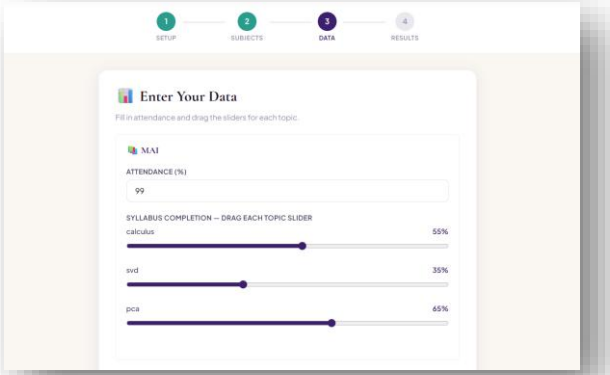
LOG IN Page



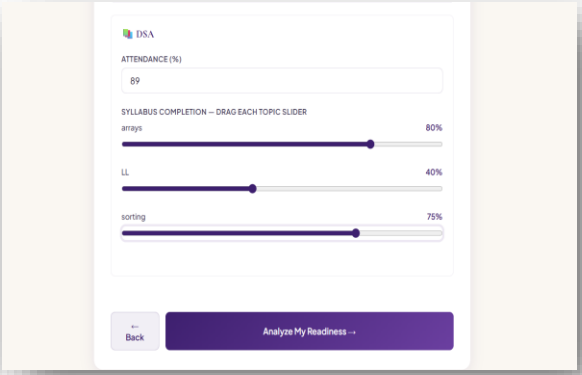
Dashboard- Exam Setup



Subject details



Subject Syllabus
& Attendance data



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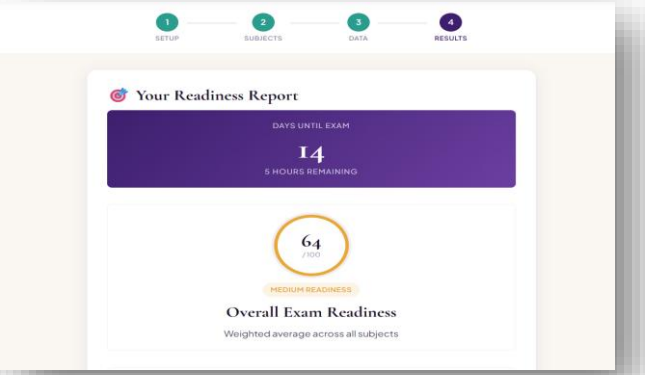
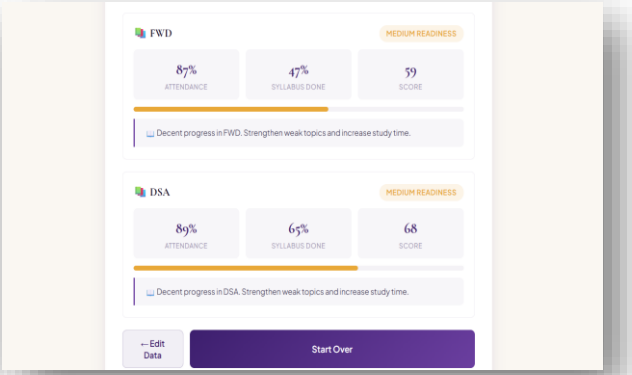


Exam Readiness
Analysis Report

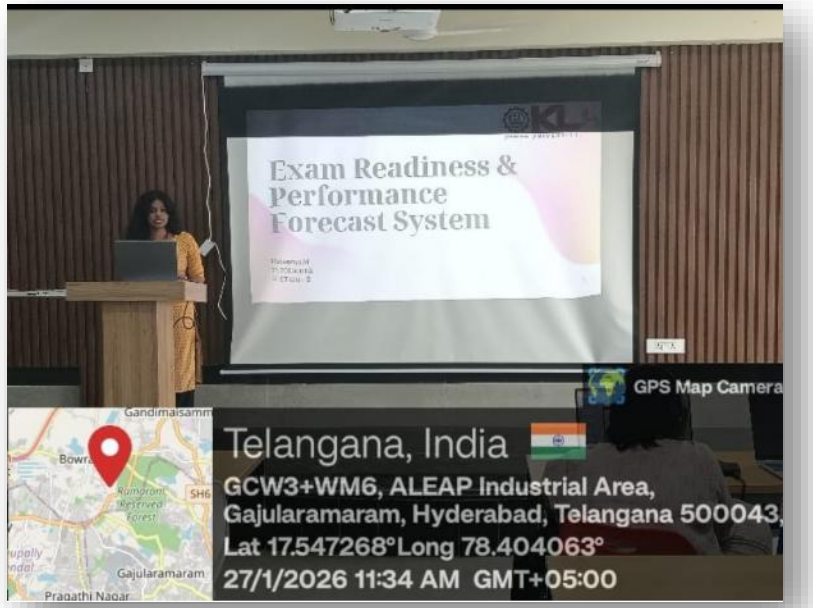


Exam Readiness
Analysis Report

Conclusion

- EduPulse demonstrates all core frontend development concepts — semantic HTML5, responsive CSS, and interactive JavaScript
- Weighted readiness formula combines three real parameters (attendance, syllabus, time) into a meaningful, actionable score
- Project is fully client-side — no frameworks, no backend, no external libraries — proving the power of core web technologies
- localStorage-based persistence enables a complete multi-session user experience without a server
- DSA concepts (Searching, Sorting, Lists, Stacks, Queues, Hashing) implemented in JavaScript in dsa/dsa.js with console output
- Responsive design works across desktop, tablet, and mobile using CSS Grid, Flexbox, and media queries
- Future scope: cloud sync for cross-device access, AI-based recommendations, peer score comparison, and mobile app version

Geo- Tag Photos



THANK YOU